Cholesky Factorization

A symmetric pos. def. $\quad A = BB^T$ for nonsingular B.

$$x^T A x > 0 \quad \forall x \neq 0$$

$A = LL^T$, L is lower triangular

<u>Ex:</u>

<u>Cholesky Factorization</u>

suppose A is SPD

From LDU factorization $\Rightarrow A = LDU$ unique L, D, U.

A sym $\Rightarrow A^T = A \Rightarrow LDU = U^T D^T L^T$

LDU is unique $\Rightarrow L = U^T$, $U = L^T$

$\Rightarrow A = LDL^T$, let $D = \mathrm{diag}(d_1, d_2, \ldots, d_n)$

consider $\vec{x}_i = (L^T)^{-1}\vec{e}_i$ = e_i is the standard basis vector

pos. def. $\Rightarrow$

$$0 < \vec{x}_i^T A \vec{x}_i = \vec{e}_i^T (L^{T^{-1}})^T L D L^T (L^T)^{-1} \vec{e}_i$$

$$= \vec{e}_i^T L^{-1} L D L^T (L^T)^{-1} \vec{e}_i$$

$$= \vec{e}_i^T D \vec{e}_i = d_i \quad \text{and all } d_i > 0$$

let $\bar{D} = \mathrm{diag}(\sqrt{d_1}, \sqrt{d_2}, \ldots, \sqrt{d_n}) \Rightarrow D = \bar{D}^2$

$$\Rightarrow A = LDL^T = L\bar{D}\bar{D}^T L^T = \underbrace{(L\bar{D})}_{\bar{L}}\underbrace{(L\bar{D})^T}_{\bar{L}^T}$$

$$\left((A^T)^{-1}\right)^T = \left((A^{-1})^T\right)^T = A^{-1} \checkmark$$

$$\left((A^T)^{-1}\right)^T \overset{?\checkmark}{=} A^{-1} \quad \text{always} \qquad \text{Yes}$$

SPD def $x^TAx > 0 \ \forall x \neq 0$

let A be an SPD matrix $\exists \{(v_k, \lambda_k)\}_{k=1}^n$ eigenpairs i.e.

$$Av_k = \lambda_k v_k \qquad \{v_k\}_{k=1}^n \text{ forms an orthonormal basis.}$$

Assume $|\lambda_1| > |\lambda_2| > \dots > |\lambda_n|$

<u>power method (I) :</u>

$$x_k = Ax_{k-1}, \qquad \mu_k = \frac{\langle Ax_k, x_k\rangle}{\langle x_k, x_k\rangle}$$

<u>power method (II) [more stable]</u>

$$w_k = Ax_{k-1}, \qquad x_k = \frac{w_k}{\|w_k\|} \qquad \mu_k = \langle Ax_k, x_k\rangle$$

A is SPD, eigenpairs $\{(\vec{v}_k, \lambda_k)\}$ $\{\vec{v}_k\}$ orthonormal

$|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$

$$\forall \vec{x}, \qquad \vec{x}_0 = \sum_{k=1}^n \alpha_k \vec{v}_k$$

$$\Rightarrow \underset{\vec{x}_1}{\underset{\parallel}{A\vec{x}_0}} = \sum_{k=1}^n \alpha_k A\vec{v}_k = \sum_{k=1}^n \alpha_k \lambda_k \vec{v}_k$$

$\vdots$ continue

$$\vec{x}_m = A^m \vec{x}_0 = \sum_{k=1}^n \alpha_k \lambda_k^m \vec{v}_k \qquad \text{if } \alpha_1 \neq 0$$

$$\frac{\vec{x}_m}{\|\vec{x}_m\|} = \frac{A^m \vec{x}_0}{\|A^m \vec{x}_0\|_{L_2}} = \frac{\sum_{k=1}^n \alpha_k \lambda_k^m \vec{v}_k}{\sqrt{\sum_{k=1}^n \alpha_k^2 \lambda_k^{2m}}}$$

$$= \frac{\sum_{k=1}^{n} \alpha_k \left(\frac{\lambda_k}{|\lambda_1|}\right)^m \vec{v}_k}{\sqrt{\sum_{k=1}^{n} \alpha_k^2 \left(\frac{\lambda_k}{\lambda_1}\right)^{2m}}} \xrightarrow{m \to \infty} \frac{\alpha_1}{|\alpha_1|} \vec{v}_1$$

$\boxed{\text{QR Iteration}}$ Find all eigenvalues simultaneously

$\in M_n(\mathbb{R})$

<u>def:</u> QR Factorization s.t. $A = QR$, Q is orthogonal $(Q^T = Q^{-1})$, and R is upper triangular.

$$A = [a_1, a_2, \dots a_n] = [q_1 \; q_2 \; \cdots \; q_n] \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ 0 & r_{22} & \cdots & r_{2n} \\ 0 & 0 & r_{33} & \cdots & r_{3n} \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & r_{nn} \end{bmatrix}$$

$$v_1 = a_1, \quad q_1 = \frac{v_1}{\|v_1\|}$$

$$v_2 = a_2 - (q_1 \cdot a_2) q_1, \quad q_2 = \frac{v_2}{\|v_2\|}$$

$$\vdots$$

$$v_k = a_k - \sum_{j=1}^{k-1} (q_j \cdot a_k) q_j, \quad q_k = \frac{v_k}{\|v_k\|}$$

<u>ex</u>

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$$

$$v_1 = \frac{1}{\sqrt{10}} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3/\sqrt{10} \\ 1/\sqrt{10} \end{pmatrix}$$

$$v_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} - \left(\begin{pmatrix} 2 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 3/\sqrt{10} \\ 1/\sqrt{10} \end{pmatrix} \right) \begin{pmatrix} 3/\sqrt{10} \\ 1/\sqrt{10} \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ 2 \end{pmatrix} - \left(6/\sqrt{10} + 2/\sqrt{10} \right) \begin{pmatrix} 3/\sqrt{10} \\ 1/\sqrt{10} \end{pmatrix} = \frac{8}{\sqrt{10}} \cdot \frac{1}{\sqrt{10}} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \frac{8}{10} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 - \frac{24}{10} \\ 2 - \frac{8}{10} \end{pmatrix} = \begin{pmatrix} -4/10 \\ +12/10 \end{pmatrix} = \begin{pmatrix} -2/5 \\ 6/5 \end{pmatrix} \qquad \sqrt{\frac{4}{25} + \frac{36}{25}} = \sqrt{\frac{40}{25}}$$

$$v_2 = \frac{5}{\sqrt{40}} \begin{pmatrix} -2/5 \\ 6/5 \end{pmatrix} = \begin{pmatrix} -2/\sqrt{40} \\ 6/\sqrt{40} \end{pmatrix} = \begin{pmatrix} -1/\sqrt{10} \\ 3/\sqrt{10} \end{pmatrix}$$

$$A = \begin{pmatrix} 3/\sqrt{10} & -1/\sqrt{10} \\ 1/\sqrt{10} & 3/\sqrt{10} \end{pmatrix} \begin{pmatrix} \cancel{a} \to \sqrt{10} & \cancel{c} \to \frac{8\sqrt{10}}{10} \\ \cancel{b} \to 0 & \cancel{d} \to \frac{4\sqrt{10}}{10} \end{pmatrix} \quad ??$$

$$\frac{3a}{\sqrt{10}} - \cancel{\frac{b}{\sqrt{10}}} = 3 \qquad b = 0$$

$$a = \frac{3\sqrt{10}}{3} = \sqrt{10} \qquad \frac{3c}{\sqrt{10}} - \frac{3d}{\sqrt{10}} = 2 \cdot 3$$

$$\frac{c}{\sqrt{10}} + \frac{3d}{\sqrt{10}} = 2$$

$$\frac{8\sqrt{10}}{10\sqrt{10}} + \frac{3d}{\sqrt{10}} = 2$$

$$8/10 + \frac{3d}{\sqrt{10}} = 2 \qquad \Rightarrow \qquad \frac{10c}{\sqrt{10}} = 8, \quad c = \frac{8\sqrt{10}}{10}$$

$$\frac{3d}{\sqrt{10}} = 12/10$$

$$d = \frac{12/10 \sqrt{10}}{3} = \frac{4\sqrt{10}}{10}$$